SKILL BUILDERS · UNIT 5 · SCORE 3

Pythagorean Theorem: Practice Set

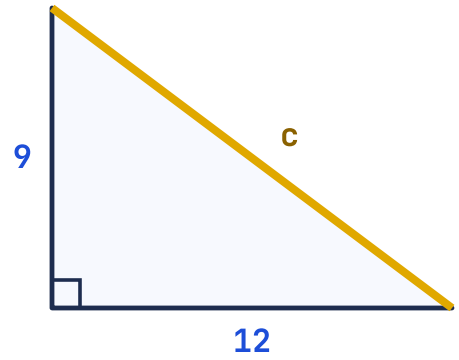
NC.8.G.6–8

Name _____ Date _____ Period _____

Do one problem at a time. Show your steps. Circle the letter or write the number.

1 Score 3 NC.8.G.7

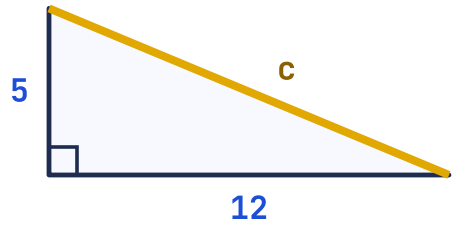
The legs of a right triangle are **9** and **12**. Find the hypotenuse.



Answer: _____

2 Score 3 NC.8.G.7

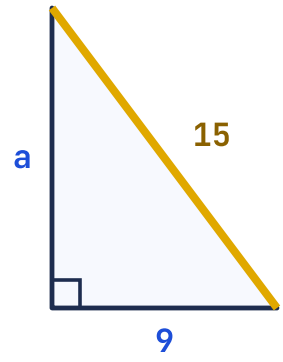
The legs of a right triangle are **5** and **12**. Find the hypotenuse.



Answer: _____

3 Score 3 NC.8.G.7

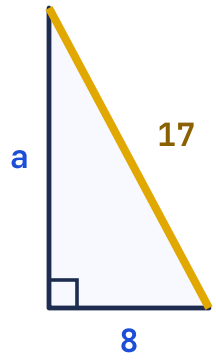
A right triangle has hypotenuse **15** and one leg **9**. Find the other leg.



Answer: _____

4 Score 3 NC.8.G.7

A right triangle has hypotenuse **17** and one leg **8**. Find the other leg.



Answer: _____

5 Score 3 NC.8.G.6

Can the sides **7, 24, 25** make a right triangle?

(A) Yes: 49 + 576 = 625, which is 25^2

(B) No: the numbers are not 3, 4, 5

(C) No: 7 + 24 is not 25

6 Score 3 NC.8.G.6

Can the sides **4, 5, 6** make a right triangle?

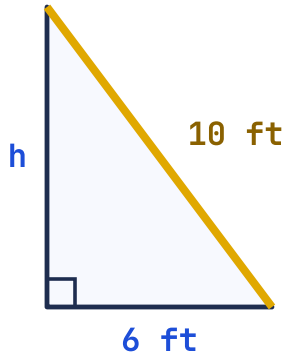
(A) No: 16 + 25 = 41, but $6^2 = 36$

(B) Yes: 4 + 5 is more than 6

(C) Yes: they are in counting order

7 Score 3 NC.8.G.7

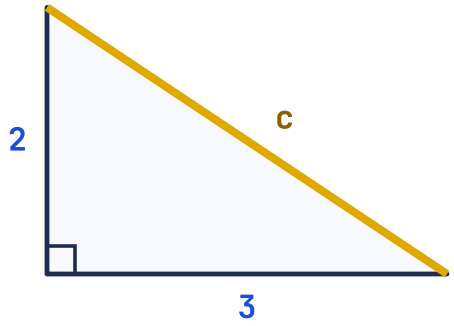
A 10-foot ladder leans on a wall. The bottom is **6 feet** from the wall. How high up the wall does it reach?



Answer: _____ ft

8 Score 3 NC.8.G.7

The legs of a right triangle are **2** and **3**. Find the hypotenuse to the nearest tenth.



Answer: _____

9 Score 3 NC.8.G.8

Find the distance from **(−2, 1)** to **(2, 4)**.

Answer: _____

10 Score 3 NC.8.G.7

Find the mistake. A right triangle has legs 6 and 8. Max wrote: **`6 + 8 = 14, so c = 14`**. What went wrong?

(A) Max is right: c = 14.

(B) Max added the legs. He must square them: 36 + 64 = 100, so c = 10.

(C) Max should subtract: 8 − 6 = 2.

(D) Max should multiply: 6 × 8 = 48.